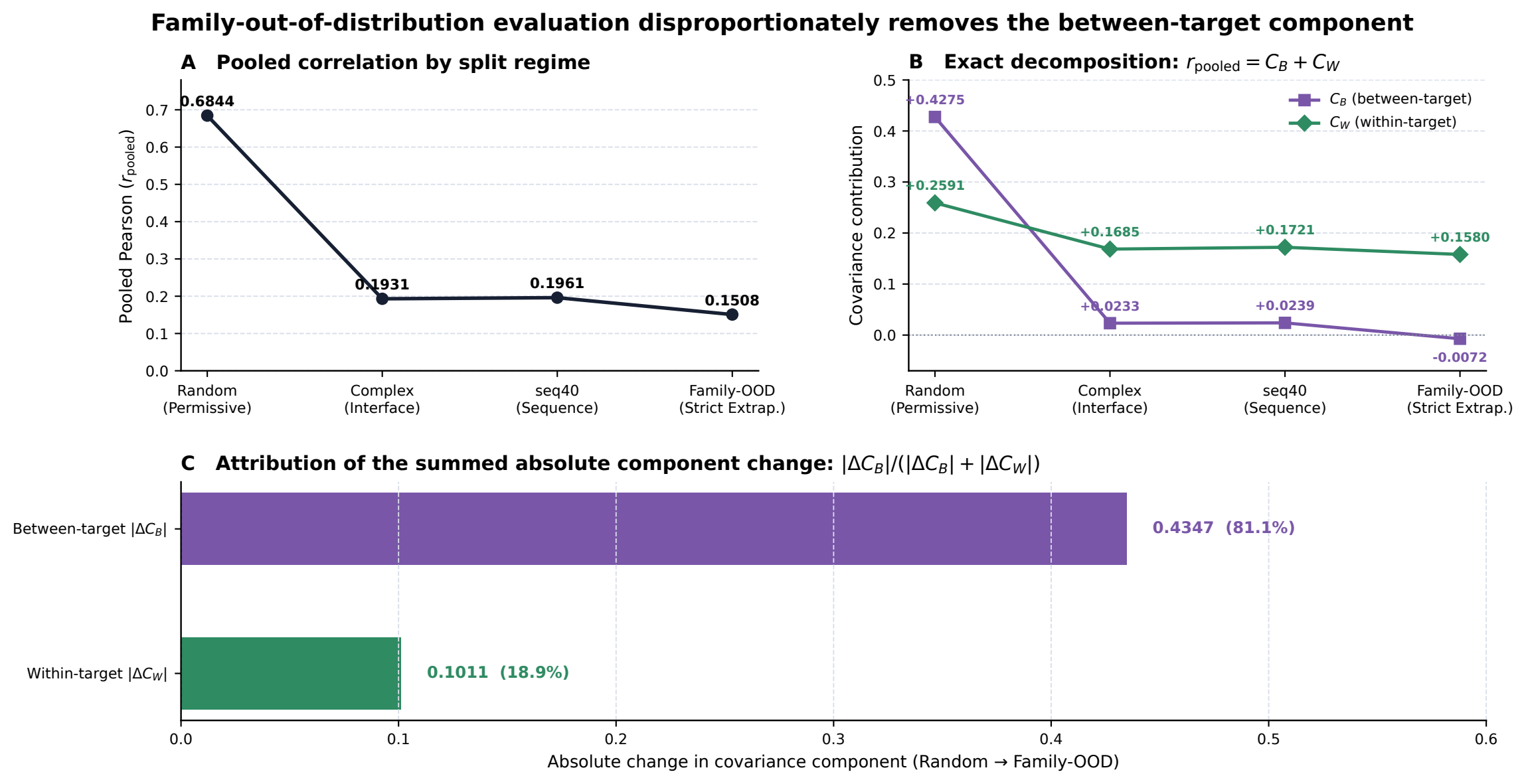
Family-out-of-distribution evaluation disproportionately removes the between-target component
A Pooled correlation by split regime
Pooled Pearson (r_{pooled})
0.6844
0.1931
0.1961
0.1508
Random (Permissive)
Complex (Interface)
seq40 (Sequence)
Family-OOD (Strict Extrap.)
B Exact decomposition: $r_{pooled} = C_B + C_W$
Covariance contribution
C_B (between-target)
C_W (within-target)
+0.4275
+0.2591
+0.1685
+0.1721
+0.1580
+0.0233
+0.0239
-0.0072
C Attribution of the summed absolute component change: $|\Delta C_B|/(|\Delta C_B| + |\Delta C_W|)$
Between-target $|\Delta C_B|$
Within-target $|\Delta C_W|$
0.4347 (81.1%)
0.1011 (18.9%)
Absolute change in covariance component (Random → Family-OOD)